

RNAGAN 1.0 Documentation (2026-03-06)

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Chapter 1. Dependency Graph



Chapter 2. File Inventory (15 files)

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DG_fine_tune.m

Size: 12.9 KB Modified: 2026-03-06 17:12

```
DG_FINE_TUNE    Train the D and G networks with samples.
    author: Zhaozheng Hou (George)

[Dnet,Gnet,Record] = DG_fine_tune(Dnet,Gnet,geneList,positiveRef,
    negativeRef,numEpochs,validationProportion,useGPU)
parameter:
    - Dnet: discriminator network to use (network or the name of
trained
    network, such as "BDPP10" for "the discriminator for bulk-RNA
seq
    data with predefined pathways and using 10 references")
    - Gnet: generator network to use (network or the name of trained
    network, such as "GDPP10" for "the generator for bulk-RNA seq
    data with predefined pathways and using 10 references"). Dnet
and
    Gnet need to have the same number of referencing data.
    - geneList: list of genes (strings or numbers), leave blank and
skip
    the matching if the data already matched the gene list
    - positiveRef: expression of positive samples, each column is one
    sample. Better with at least 20 samples.
    - negativeRef: expression of positive samples, each column is one
    sample. Better with at least 20 samples.
    - numEpochs: number of epochs (how many times each sample get used
for
    training). At least 10 epochs.
    - validationProportion: (optional) proportion of first samples
used for
    validation, 0.3 by default. 0 for training without validation,
1
    for only validate the network without training.
    - useGPU: (optional) whether using GPU for the processing or not,
true
    by default
output:
    - Dnet: tuned discriminator network
    - Gnet: tuned discriminator network
    - Record: structure of training records: recoreded epochs,
D/Gnetworks,
    D/Gvalidations, and training scores and loss
```

D_diagnose.m

Size: 5.1 KB Modified: 2026-03-03 21:35

```
D_DIAGNOSE    Diagnose the given target is the same as the positive
               reference or not.
               author: Zhaozheng Hou (George)

P = D_diagnose(Dnet, geneList, target, positiveRef, useGPU, negativeRef)
parameter:
- Dnet: discriminator network to use (network or the name of
trained
      network, such as "BDPP10" for "the discriminator for bulk-RNA
seq
      data with predefined pathways and using 10 references")
- geneList: list of genes (strings or numbers), leave blank and
skip
      the matching if the data already matched the gene list
- target: expression of targets, each column is one sample
- positiveRef: expression of positive samples, each column is one
sample. Better with at least 20 samples.
- useGPU: (optional) whether using GPU for the processing or not,
false
      by default
- negativeRef: (optional) expression of positive samples, each
column
      is one sample
output:
- P: Probability of the target is of the same type as the positive
```

D_extract_eatures.m

Size: 16.4 KB Modified: 2026-03-06 17:13

D_EXTRACT_EATURES Extract features of given feature order.

author: Zhaozheng Hou (George)

```
Features = D_extract_eatures(Dnet, geneList, positiveRef, negativeRef,  
                             featureOrder)
```

parameter:

- Dnet: discriminator network to use (network or the name of
trained network, such as "BDPP10" for "the discriminator for bulk-RNA
seq data with predefined pathways and using 10 references")
 - geneList: list of genes (strings or numbers), leave blank and
skip the matching if the data already matched the gene list
 - positiveRef: expression of positive samples, each column is one
sample. Better with at least 20 samples.
 - negativeRef: expression of positive samples, each column is one
sample. Better with at least 20 samples.
 - featureOrder: order of features to extract. 0/1/2
- output:
- Features: structure of extracted features

D_fine_tune.m

Size: 9.7 KB Modified: 2026-03-06 17:12

```
D_FINE_TUNE    Train the network (D only) with samples
               author: Zhaozheng Hou (George)

[Dnet,Record] =
D_fine_tune(Dnet,geneList,positiveRef,negativeRef,echo,
            validationProportion,useGPU)
parameter:
- Dnet: discriminator network to use (network or the name of
trained
      network, such as "BDPP10" for "the discriminator for bulk-RNA
seq
      data with predefined pathways and using 10 references")
- geneList: list of genes (strings or numbers), leave blank and
skip
      the matching if the data already matched the gene list
- positiveRef: expression of positive samples, each column is one
sample. Better with at least 20 samples.
- negativeRef: expression of positive samples, each column is one
sample. Better with at least 20 samples.
- numEpochs: number of epochs (how many times each sample get used
for
      training). At least 10 epochs.
- validationProportion: (optional) proportion of first samples
used for
      validation, 0.3 by default. 0 for training without validation,
1
      for only validate the network without training.
- useGPU: (optional) whether using GPU for the processing or not,
true
      by default
output:
- Dnet: tuned discriminator network
- Record: structure of training records: recoreded epochs,
networks,
      validations, and training scores and loss
```

G_generate_pseudo.m

Size: 3.6 KB Modified: 2026-03-06 17:09

G_GENERATE_PSEUDO Generate pseudo data according to the given references.

author: Zhaozheng Hou (George)

```
[pseudo,pseudoNet] = G_generate_pseudo(Gnet, geneList, reference,
    numberOfPseudo, useGPU)
```

parameter:

- Gnet: generator network to use (network or the name of trained network, such as "BGPP10" for "the generator for bulk-RNA seq data with predefined pathways and using 10 references")
 - geneList: list of genes (strings or numbers), leave blank and skip the matching if the data already matched the gene list
 - reference: expression of smples, each column is one sample. At least 2 samples.
 - numberOfPseudo: number of pseudo samples to generate, 1 by default.
 - useGPU: (optional) whether using GPU for the processing or not, false by default.
- output:
- pseudo: data with the same gene list as reference, each column is one pseudo sample. (unmatched genes get NA)
 - pseudoNet: data with the gene list same as the network.

G_vectorize.m

Size: 3.9 KB Modified: 2026-03-06 17:09

```
G_VECTORIZE    Vectorize the reference sample group into 64
dimensional
    vector(s). The vector is for the entire group because, because the
    group variance of genes are also considered in vectorization.
    user can also use a numberOfVectors larger than 1 and the vector
would
    be generated by random resampling.
    author: Zhaozheng Hou (George)

V = G_vectorize(Gnet, geneList, reference, numberOfVectors, useGPU)
parameter:
    - Gnet: generator network to use (network or the name of trained
        network, such as "BGPP10" for "the generator for bulk-RNA seq
        data with predefined pathways and using 10 references")
    - geneList: list of genes (strings or numbers), leave blank and
skip
        the matching if the data already matched the gene list
    - reference: expression of smpls, each column is one
        sample. At least 2 samples.
    - numberOfVectors: number of vectors to generate, 1 by default.
    - useGPU: (optional) whether using GPU for the processing or not,
false
        by default.
output:
    - V: each column is one vectorized result
```

export_network_as_TF.m

Size: 4.0 KB Modified: 2026-03-06 17:10

EXPORT_NETWORK_AS_TF Export a network as a Tensorflow model
author: Zhaozheng Hou (George)

export_network_as_TF(network,pythonPackageName)

parameter:

- network: network, or name of the network
- pythonPackageName: packagename, also the folder name

get_network_info.m

Size: 0.9 KB Modified: 2026-03-06 17:03

GET_NETWORK_INFO display the structure and input size of a deep learning network.

author: Zhaozheng Hou (George)

```
inputSize = get_network_info(network)
```

parameter:

- net: the network to analysis

output:

- inputSize: the input size of the network

match_gene_list.m

Size: 4.6 KB Modified: 2026-01-18 15:13

MATCH_GENE_LIST match a given gene list to the list for RNAGAN network.

Only the unique matches would remain.

author: Zhaozheng Hou (George)

Example

```
network(netGeneID)=data(listGeneID);
network(netGeneID)=data(matched);
data(listGeneID)=network(netGeneID);
```

```
[netGeneID,listGeneID,matched] = match_gene_list(geneList, format)
```

parameter:

- geneList: the list of genes (strings or numbers)
- format: (optional) the format of gene list:

"ENSG"/"Entrez"/"symbol"

output:

- netGeneID: the IDs of matched genes in net input layer
- listGeneID: the IDs of matched genes in gene list
- matched: bool array telling which gene on the list got unique

match

visualize_2order_features.m

Size: 2.3 KB Modified: 2026-03-06 17:09

```
VISUALIZE_2ORDER_FEATURES   Generate plot the 2nd order features  
author: Zhaozheng Hou (George)
```

```
[pseudo,pseudoNet] = visualize_2order_features(gradients,R2,  
markerNames,selectedMarkers)
```

parameter:

- gradients: gradient matrix of features
- R2: partial R2 values fo the features
- markerNames: (optional) marker names, feature order by default.
- selectedMarkers: (optional) features to show, show all by

default.

output:

- P: plot object
- G: graph object

core\dataSplitLayer.m

Size: 0.5 KB Modified: 2026-03-06 17:18

```
split the data of a layer
```

core\modelLossD.m

Size: 0.4 KB Modified: 2026-03-06 17:17

```
compute the loss for Discriminator
```

core\modelLossDwG.m

Size: 0.6 KB Modified: 2026-03-06 17:17

```
compute the loss for Discriminator against Generator
```

core\modelLossG.m

Size: 0.4 KB Modified: 2026-03-06 17:16

```
compute the loss for Generator
```

core\reshapeLayer.m

Size: 0.6 KB Modified: 2026-03-06 17:20

```
layer for reshaping (permute in dimensions) the data
```